OENG1209: Engineering Science

Week 4:

Rotational Motion

RMIT UNIVERSITY

Today's Topics

- Uniform Circular Motion
- Angular Displacement and Velocity.
- Accelerations in Circular Motion:
  - Centripetal Acceleration .
  - Angular Acceleration.
- Dynamics of Circular Motion

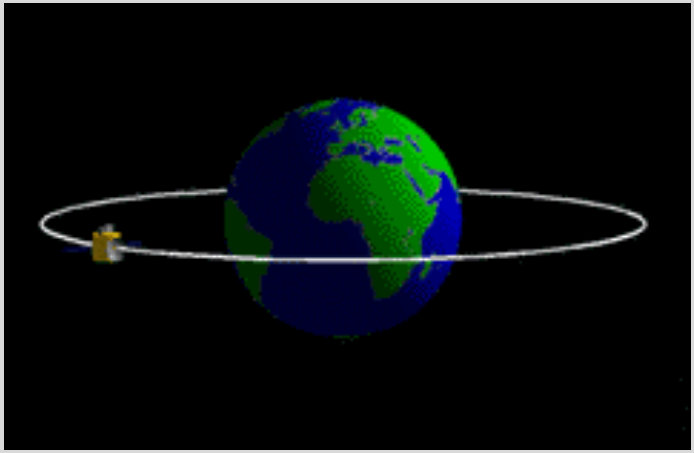
Circular Motion

Round and Round they go

RMIT UNIVERSITY

Uniform circular motion

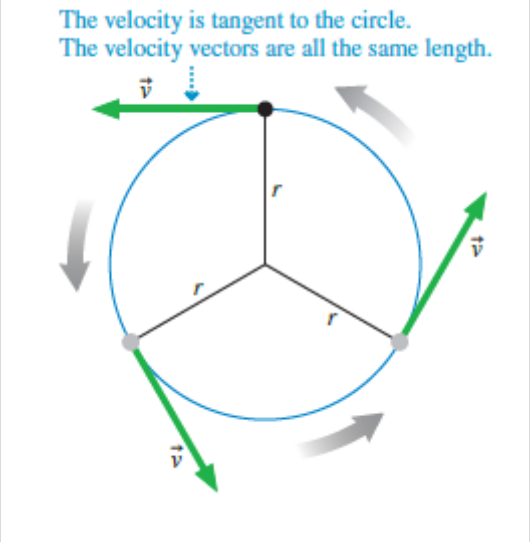
- Consider the motion of a satellite in orbit around the Earth.
- How do we calculate its **position**, **velocity** and **acceleration**?



Uniform circular motion

- We simplify the problem and reduce the satellite to a single particle (a point) that moves at a constant speed around a circle of **radius** r.
- The particle's speed is constant, so **vector** $\vec{v}$ is always the same length.

magnitude doesn't change
But direction change



$$\vec{V} = \frac{d\vec{r}}{dt} = \lim_{\Delta t \to 0} \frac{\Delta r}{\Delta t}$$



as $\Delta t \to 0 \to \alpha = 90°$

when $\Delta t \to 0$
r and $= r + \Delta r$



$\alpha' = \frac{180-\beta}{2}$

$\frac{180-0}{2} = 90°$

so $\Delta r \to 0 \to \beta \to 0 \to \alpha \to 90°$

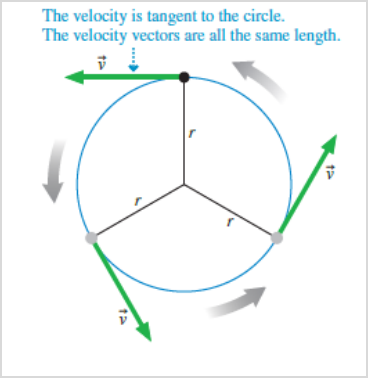
Uniform circular motion

- Recall **the Circumference** of a circle is given by:

$$C = 2\pi r$$

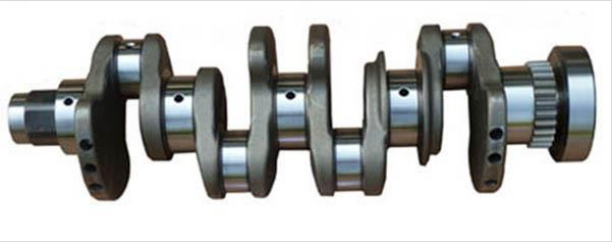
- The time it takes the particle to go one full circle is defined as **a period T**.
- Hence the speed of a particle per 1 period is:

$$v = 1\frac{circumference}{1\ period} = \frac{2\pi r}{T}$$



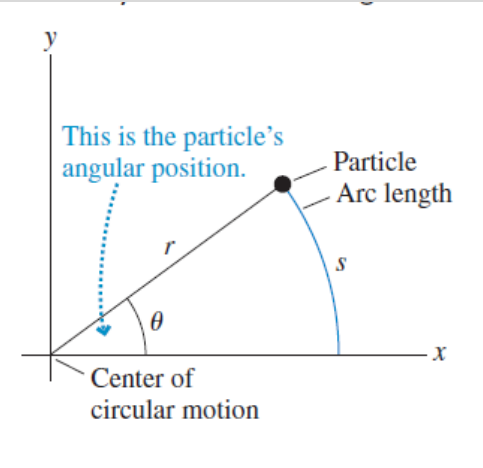
Uniform circular motion – Example

- **Example:** A 4.0 cm diameter crankshaft turns at 2400 revolutions per minute. What is the speed of a point on the surface of the crankshaft?



Angular Position

- When discussing the angular position of an object, it is more convenient to look at its **distance r** from the center of the circle and the **angle θ** from the positive $x-$ axis.
- **Counterclockwise** (ccw) motion is positive and **Clockwise** (cw) motion is negative.



Angular Position

- **The angle θ** is used as the **angular position** of a particle, expressed in **radians** (not degrees).
- **The angle θ** is given by finding the **ratio** between the arc-length of the circle s and the radius of the circle r

$$\theta = \frac{s}{r}$$

- Hence the total length of the path of a particle in circular motion is:

$$s = \theta r$$

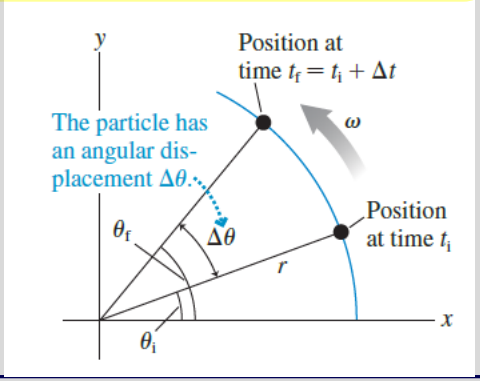
Angular Velocity

- Similar to linear displacement, **the angular displacement** of an object is defined as:

$$\Delta\theta = \theta_f - \theta_i$$

- Hence, we can define the **average angular velocity** for an object to be:

$$average\ angular\ velocity = \frac{\Delta\theta}{\Delta t}$$

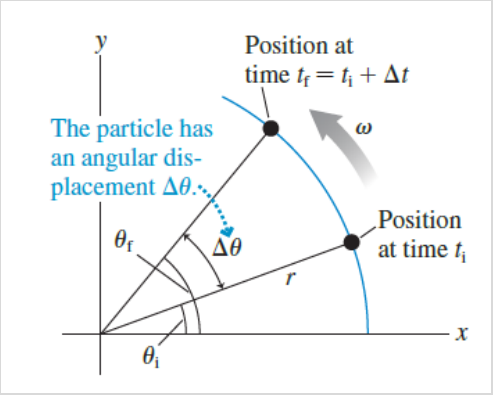


Angular Velocity

- Similar to finding velocity by taking a limit of $\Delta t \to 0$, we can have the definition of **instantaneous angular velocity ω** as:

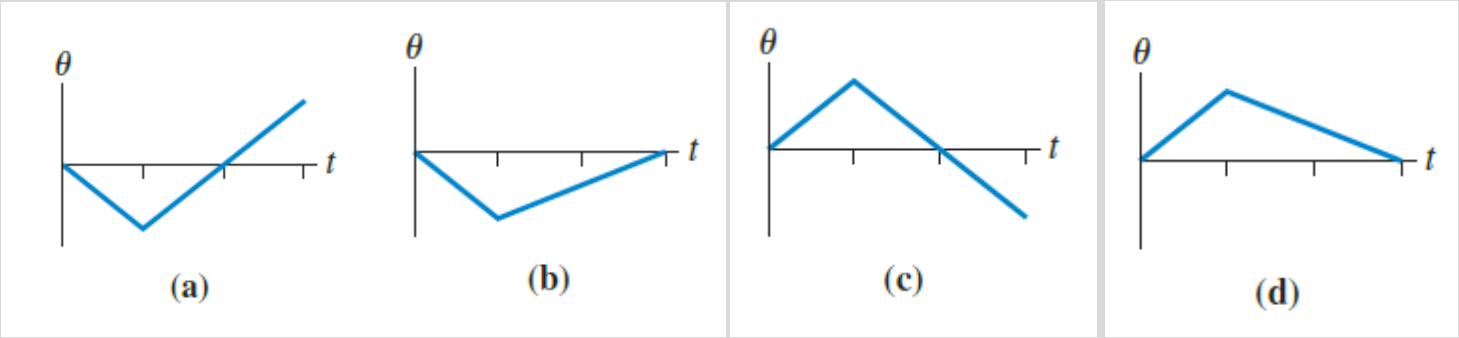
$$\omega = \lim_{\Delta t \to 0} \left(\frac{\Delta \theta}{\Delta t} \right) = \frac{d\theta}{dt}$$

- Angular velocity has the unit of ***rad/second*** or ***revolution/second***.



Angular Velocity – Example

- **Example:** A particle moves clockwise around a circle at constant speed for 2.0 s. It then reverses direction and moves counter-clockwise at half the original speed until it has traveled through the same angle. Which one is the particle's angle-versus-time graph?

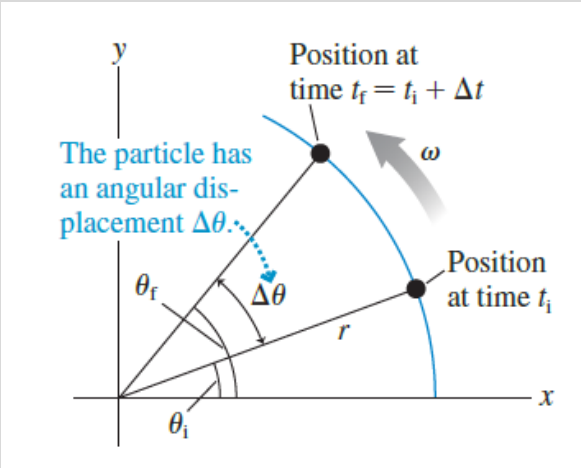


Angular Velocity

- The relationship between **angular velocity** and **angular position** is similar to the relationship between linear velocity and position:

$$x_f = x_i + vt$$

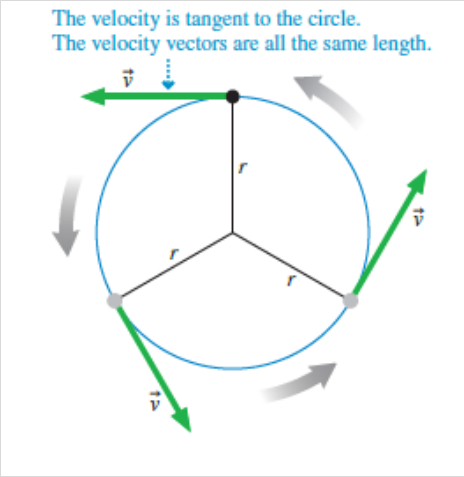
$$\rightarrow \boldsymbol{\theta_f = \theta_i + \omega t}$$



Angular Velocity

- This **velocity vector** $\vec{v}$ is called the **tangential velocity**, which is positive for ccw motion and negative for cw motion.
- The **tangential velocity vector** $\vec{v}$ is given by:

$$v_t = \omega r$$



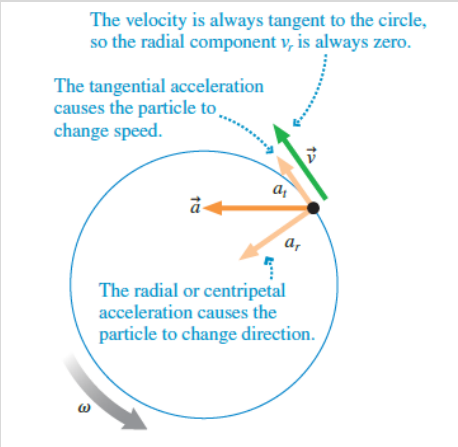
Angular Velocity – Example

- **Example:** A small steel roulette ball rolls ccw around the inside of a 30 cm diameter roulette wheel. The ball completes 2.0 rev in 1.20 seconds. (a) What is the ball's angular velocity? (b) What is the ball's position at $t = 2.0\ s$? (Assume $\theta_i = 0$)



Circular Acceleration

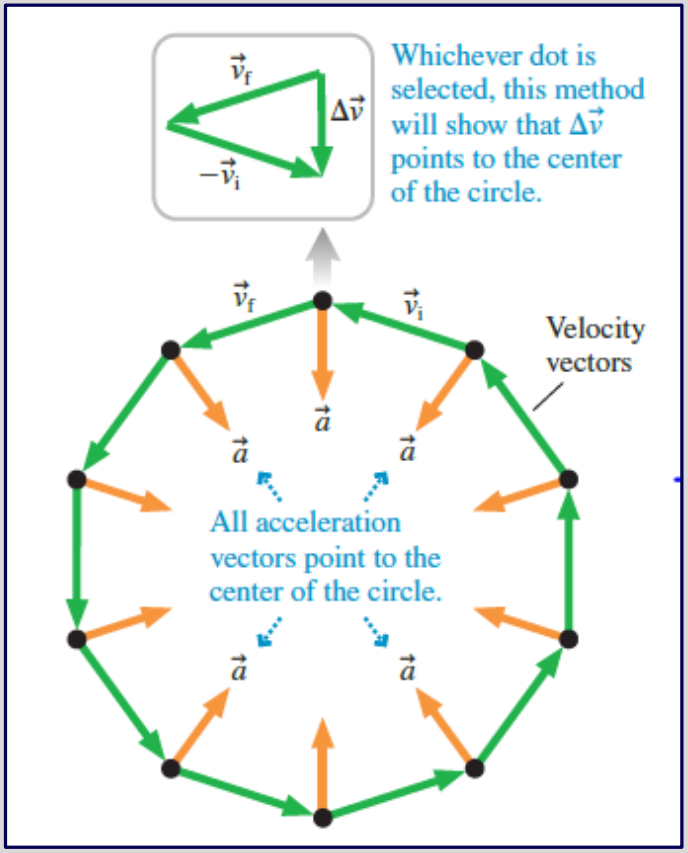
- There are **2 components** to the acceleration of a particle in circular motion: centripetal acceleration, and tangential acceleration.
- Centripetal acceleration exists for uniform and non uniform circular motion.
- Tangential acceleration exists for non-uniform circular motion.



Centripetal Acceleration

- **Centripetal Acceleration** is also called the radial acceleration.
- Recall the definition of acceleration as the **change in magnitude or direction of the velocity vector**.
- Observe that as the object is moving around a circle, its velocity vector is constantly changing its direction (while magnitude remains constant).

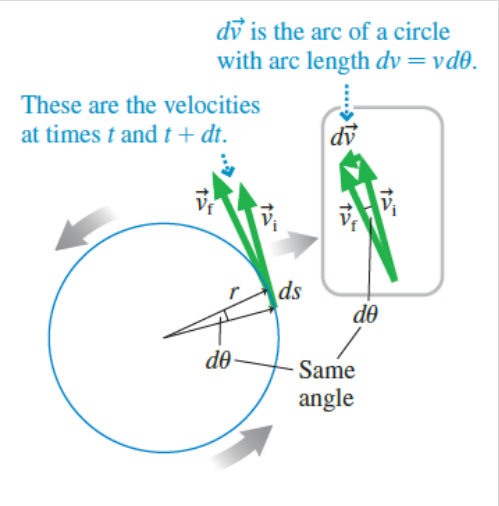
Centripetal Acceleration



Centripetal Acceleration

- Observe that as the object moves by a $d\theta$ displacement, its tangential velocity vector is changed by an amount of $d\vec{v}$.
- Hence we can approximate $d\vec{v}$ as **the arc of a circle** with arc length:

$$dv = vd\theta$$



Centripetal Acceleration

- For uniform circular motion at constant speed, the tangential velocity is:

$$|\vec{v}| = \frac{ds}{dt} = \frac{rd\theta}{dt}$$

$$\rightarrow \boldsymbol{dt = \frac{rd\theta}{v}}$$

- Recall the definition of acceleration, we have:

$$|\vec{a}_c| = \frac{|d\vec{v}|}{dt} = \frac{vd\theta}{\frac{rd\theta}{v}} = \frac{v^2}{r}$$

Centripetal Acceleration

- Hence, the centripetal acceleration vector has the direction **always pointing to the center of the circle of motion** with magnitude:

$$|\vec{a_c}| = \frac{v^2}{r}$$

- Recall that $v = \omega r$, we can also write the magnitude of the centripetal acceleration as:

$$|\vec{a}| = \omega^2 r$$

- **Important:** the centripetal acceleration is not constant as its direction is always changing (pointing toward the center of the circle).

Centripetal Acceleration – Example

- **Example:** A typical Ferris Wheel has a radius of 9.0 m and rotates 4.0 times per minute. What speed and acceleration do the riders experience?



Angular Acceleration

- The angular acceleration of an object in circular motion is defined as the **change in the magnitude of its angular velocity**. The angular acceleration is denoted as α (alpha).

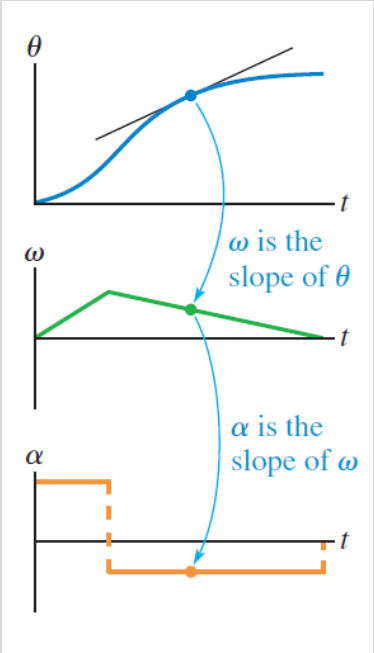
$$\alpha = \frac{d\omega}{dt}$$

- The unit of angular acceleration is rad/s^2.

Angular Acceleration – Kinematic Equations

- If the angular acceleration is constant, we can apply the same 3 kinematic equations discussed previously to solve angular displacement problems.
- **Analogs:** $s \rightarrow \theta$, $v_s \rightarrow \omega$, $a_s \rightarrow \alpha$

Rotational kinematics	Linear kinematics
$\omega_f = \omega_i + \alpha \Delta t$	$v_{fs} = v_{is} + a_s \Delta t$
$\theta_f = \theta_i + \omega_i \Delta t + \frac{1}{2}\alpha(\Delta t)^2$	$s_f = s_i + v_{is} \Delta t + \frac{1}{2}a_s(\Delta t)^2$
$\omega_f^2 = \omega_i^2 + 2\alpha \Delta\theta$	$v_{fs}^2 = v_{is}^2 + 2a_s \Delta s$



Angular Acceleration – Example

- **Example:** A ceiling fan spinning at 60 rpm coasts to a stop 25 s after being turned off. How many revolutions does it make while stopping?



Total Acceleration for Circular Motion

- As discussed previously, the acceleration of an object in Circular Motion has 2 components: **Centripetal Acceleration** and **Tangential Acceleration**.
- **The Centripetal Acceleration** is given by :

$$a_C = \frac{v^2}{r}$$

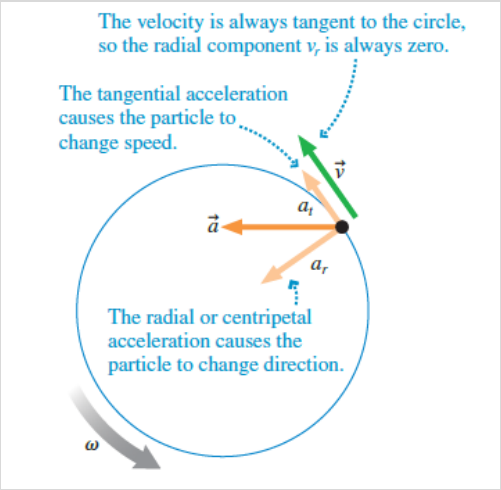
- **The Tangential Acceleration** is given by the Angular Acceleration (analogous to the Tangential velocity and Angular velocity).

$$a_T = \alpha r$$

Total Acceleration for Circular Motion

- Hence the magnitude of the total acceleration for an object in Circular Motion is:

$$a_{total} = \sqrt{a_C^2 + a_T^2}$$



Dynamics of Uniform Circular Motion

The Central Force Model

RMIT UNIVERSITY

Dynamics of Uniform Circular Motion

- Since an object travelling in **uniform circular motion** is constantly changing its velocity, thus according to **Newton's First Law**, there must be **a net force** acting on it.



Dynamics of Uniform Circular Motion

- Recall the definition of **the centripetal acceleration**:

$$|\vec{a}_c| = \frac{v^2}{r} = \omega^2 r$$

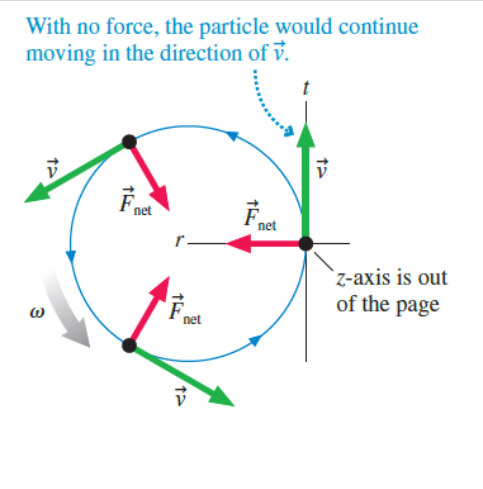
- Thus, we can invoke Newton's Second Law to quantify this **"centripetal" net force**:

$$\vec{F}_{net} = m\vec{a} = m\frac{v^2}{r}$$

$$(toward\ the\ center\ of\ the\ circle)$$

Dynamics of Uniform Circular Motion

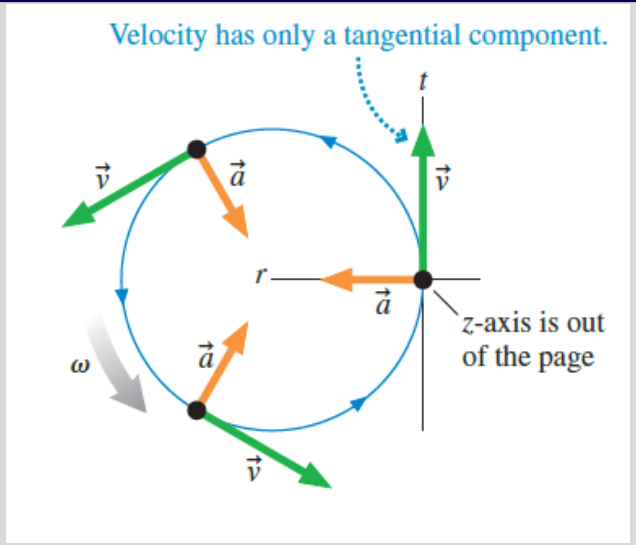
- Any object moving in a circle must have a net force with magnitude equals to $\frac{mv^2}{r}$ pointing toward the center of the circle. Otherwise, the object will move off in the tangential direction of vector $\vec{v}$.

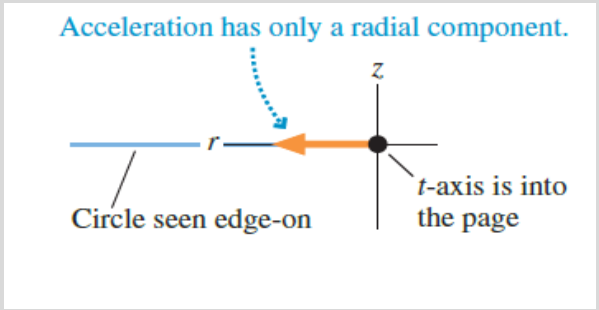


Dynamics of Uniform Circular Motion – *rtz* Coordinate System

- With the **tangential** and **radial direction** of motion, we can see that the **2D Cartesian Coordinate** system is quite inconvenient to convey the net forces for objects in Circular motions.
- Hence, for angular displacement problems, we shall use the rtz coordinate system instead of the xyz-coordinate system.
- The $\boldsymbol{r}$ stands for radial and $\boldsymbol{t}$ stands for tangential axis.

Dynamics of Uniform Circular Motion





Dynamics of Uniform Circular Motion

- Hence, instead of finding the net force in the xyz directions, we will find the **net force in the rtz directions**.
- For uniform circular motion, the summation of Forces according to Newton's 2nd Law in the radial, tangential, and z-directions shall be:

$$\sum F_r = ma_r = \frac{mv^2}{r} = m\omega^2 r$$

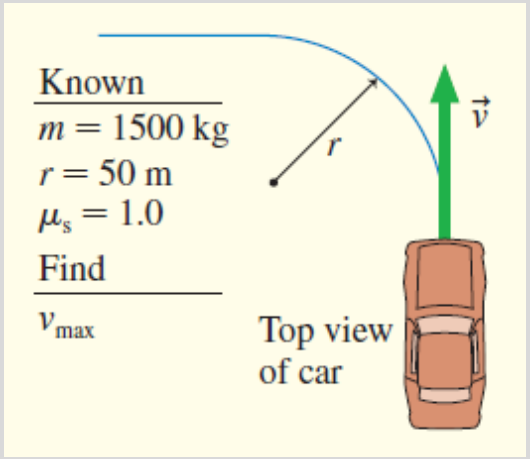
$$\sum F_t = ma_t = 0$$

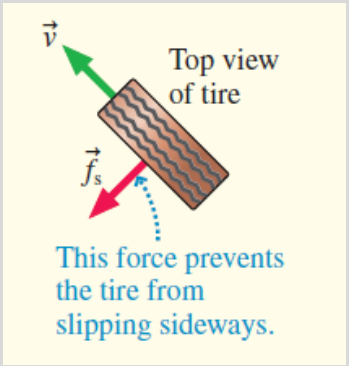
$$\sum F_z = ma_z = 0$$

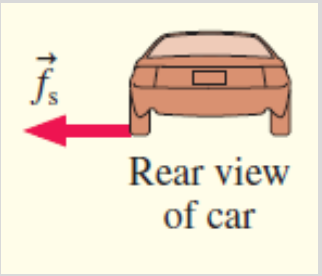
- This is called the **Central – Force Model**.

Uniform circular motion – Example 1

- **Example:** What is the maximum speed with which a 1500 kg car can make a left turn around a curve of radius 50 m on a level (unbanked) road without sliding? The coefficient of static friction between the tire and the road is $\mu_S = 1.0$

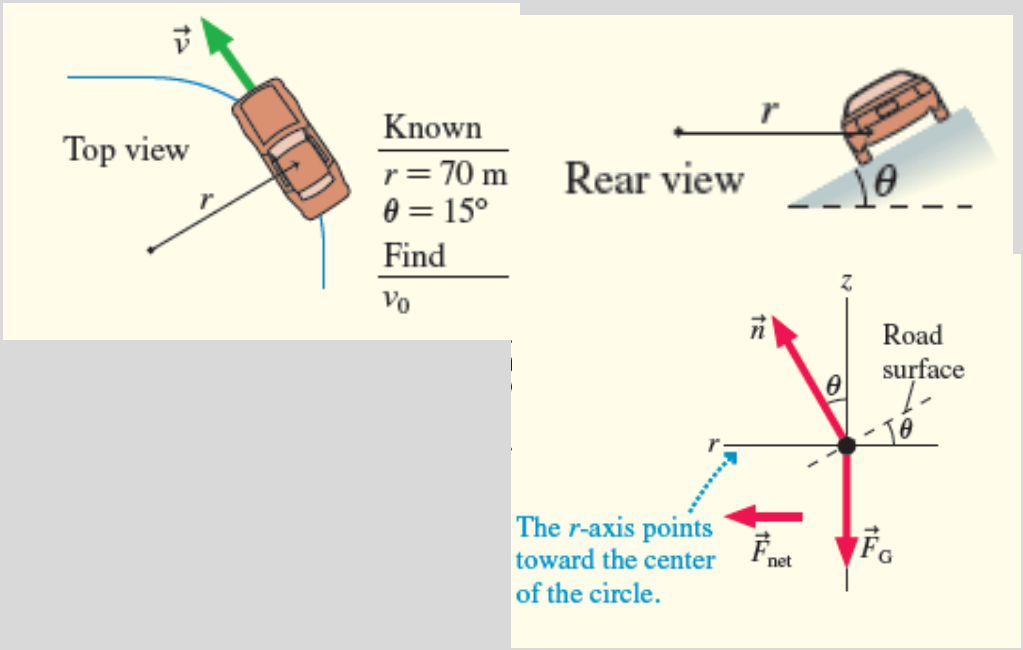






Uniform circular motion – Example 2

- **Example:** A highway curve of radius 70 m is banked at a 15° angle. At what speed v_0 can a car take this curve from friction?



Uniform circular motion – Example 2

- What happens if we increase θ to 90°?



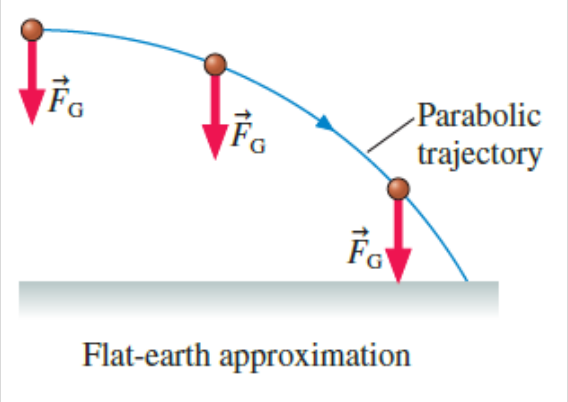
Centripetal Acceleration applications

Satellites and Gravity on Earth

RMIT UNIVERSITY

Circular Orbit

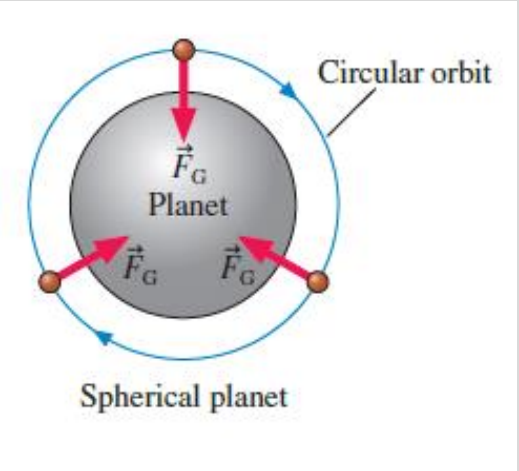
- **Centripetal Acceleration** is the main reason to why **circular orbit** is possible.
- Recall the example of a satellite orbiting the Earth, notice that it is constantly being pulled toward the Earth by the **gravitational acceleration g**. If the Earth was flat, the satellite will crash.



Circular Orbit

- However, since the Earth is approximately **spherical**, the satellite is actually under the pulling force in the direction toward the Center of the Earth.
- This acceleration is:

$$a_r = \frac{v_{orbit}^2}{r} = g$$



Circular Orbit

- Thus, if an object wants to maintain a **perfectly circular orbit around the Earth**, it must be moving at the speed of at least:

$$v_{orbit} = \sqrt{rg}$$

r is the distance to the center of the Earth.

g is the gravitational acceleration.

- In this orbit, the satellite is constantly "falling" toward the Earth but never gets any closer to the surface.

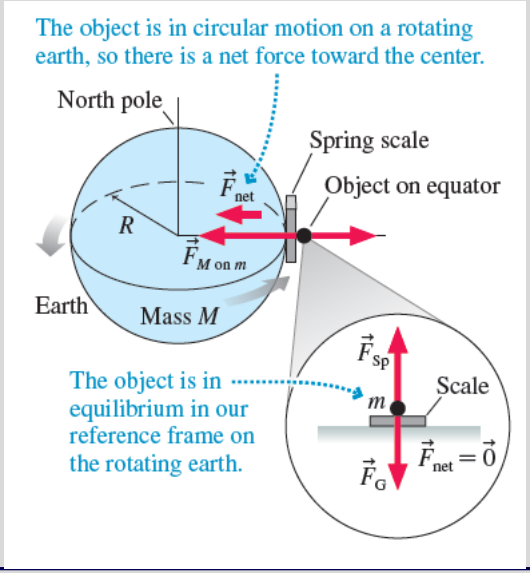
Circular Orbit – Example 1

- **Example:** Suppose a satellite is launched into orbit at $h = 3.8 \times 10^5 m$ above the Earth's surface. How fast will the satellite needs to go to maintain its orbit and what is its period? The Earth's radius is $r = 6.37 \times 10^6\ m$



Gravity on a Rotating Earth

- Consider an object being laid on a scale at the Earth's equator.
- An astronaut from outer space will observe that there is actually a centripetal force pushing the object toward the Earth's center in addition to Gravity.



Gravity on a Rotating Earth

- Hence, in the radial direction, Newton 2nd Law will give us:

$$\sum \boldsymbol{F_r = F_{Earth\ on\ m} - F_{SP} = m\omega^2 R}$$

Where **R** is the Radius of the Earth and $\boldsymbol{\omega}$ is the Earth's angular speed.

- Hence the total Force that the Earth exerts on the object is:

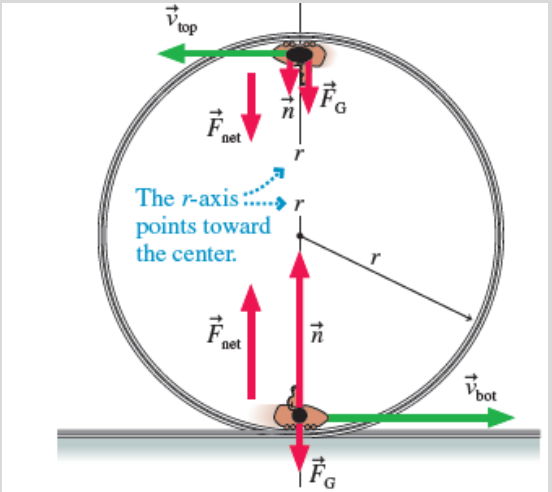
$$\boldsymbol{F_{Earth\ on\ m} = F_{SP} + m\omega^2 R}$$

- What does $\omega^2 R$ equals to if $\omega = 1\ rev/day$ and $R_{Earth} = 6370\ km$?

Why does the car stay on track?

- Consider a roller coaster going through a loop-the-loop, **why don't the car fall to the ground?**
- A Free-body-diagram of the situation can be drawn as below:





Why does the car stay on track?

- If we invoke Newton's 2nd law in the radial direction when the car is **at the bottom of the loop**, we will find that:

$$\Sigma F_r = m\, a_r = N_{bot} - mg = \frac{mv^2}{r}$$

$$\rightarrow \boldsymbol{N_{bot} = mg + \frac{mv^2}{r}}$$

- This indicates that **the normal force** at the bottom of the loop is larger than the weight of the car.

Why does the car stay on track?

- Next, we invoke Newton's 2nd law in the radial direction when the car is **at the top of the loop**, we will find that:

$$\Sigma F_r = m\, a_r = N_{top} + mg = \frac{mv^2}{r}$$

$$\rightarrow \boldsymbol{N_{top} = \frac{mv^2}{r} - mg}$$

- This indicates that there exists a **small normal force** pushing the car to the center of the track if $\frac{v^2}{r} > g$.
- If N_{top} is zero, there is no contact between the car and the track and the car falls down.

Why does the car stay on track?

- The speed of the roller coaster at which $N_{top} = 0$ is called **the critical speed v_c**:

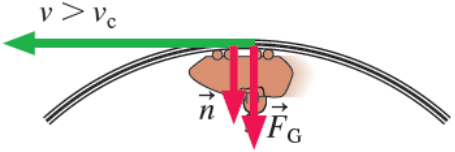
$$v_c = \sqrt{\frac{rmg}{m}} = \sqrt{rg}$$

- Below this speed, there is no normal force and no contact between the car and the track. Thus, the car will fall to the ground.
- Note that v_c is the same as v_{orbit}.

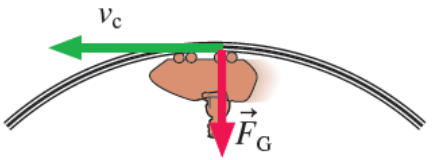
Why does the car stay on track?

- The importance of the **critical speed v_c** is illustrated below:

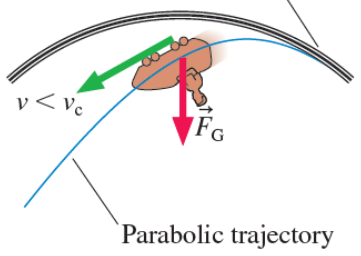
The normal force adds to gravity to make a large enough force for the car to turn the circle.



At v_c, gravity alone is enough force for the car to turn the circle. $\vec{n} = \vec{0}$ at the top point.

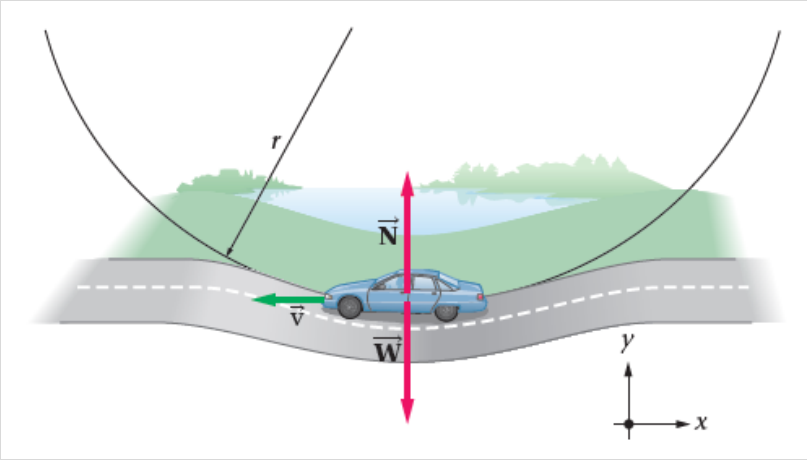


The gravitational force is too large for the car to stay in the circle!



Normal Force in a dip – Example

- **Example:** While driving along a country lane with a constant speed of 17.0 m/s, you encounter a dip in the road. The dip can be approximated as a circular arc, with a radius of 65.1 meters. What is the normal force exerted by a car seat on an 80 kg passenger when the car is at the bottom of the dip?



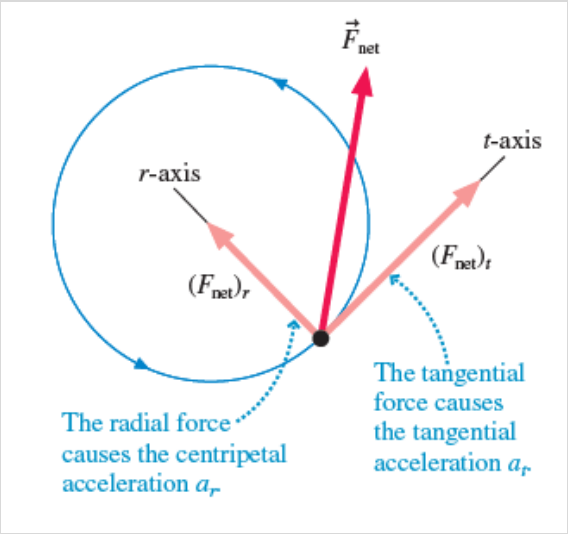
Nonuniform Circular Motion

Satellites and Gravity on Earth

RMIT UNIVERSITY

Dynamics of Nonuniform Circular Motion

- In **nonuniform circular motions**, we have the object speeding up in the tangential direction of its circular path. Hence, it's tangential acceleration will be:



$$a_t = \frac{dv_t}{dt}$$

Dynamics of Nonuniform Circular Motion

- For Nonuniform Circular Motion, Newton's 2nd Law stills hold, but the summation of forces in the tangential direction is no longer zero:

$$\sum F_r = ma_r = \frac{mv^2}{r} = m\omega^2 r$$

$$\sum F_t = ma_t$$

$$\sum F_z = ma_z = 0$$

- Remember that the convention for tangential force/acceleration/velocity is **positive for ccw** and **negative for cw**.

Dynamics of Nonuniform Circular Motion – Example

- **Example:** A 1500 kg car drives around a flat, 50 m diameter track, starting from rest. The engine supply a steady 525 N force in the forward direction. The coefficient of static friction between the car tires and the road is $\mu_S = 0.9$. How many revolutions of the track will be made before the car slides out of the curve?

